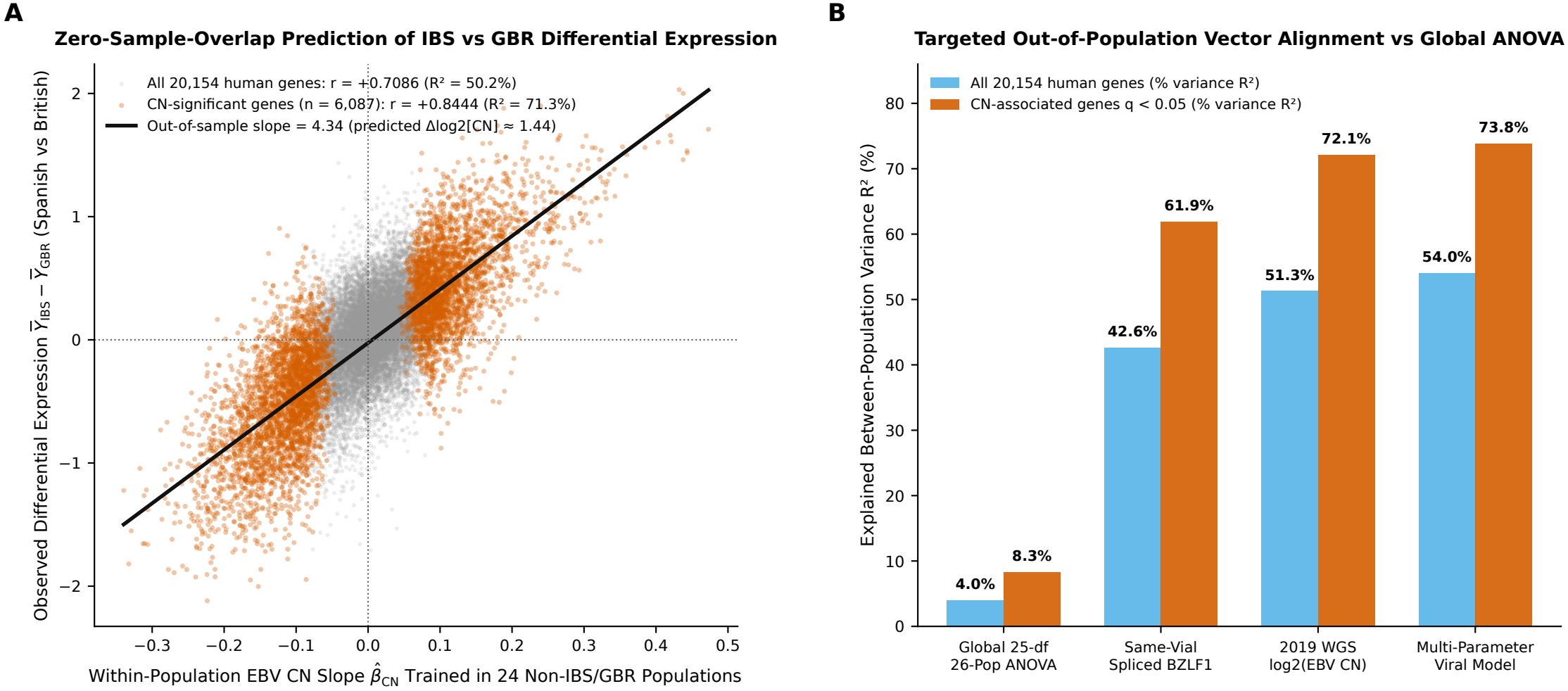

A
Zero-Sample-Overlap Prediction of IBS vs GBR Differential Expression
All 20,154 human genes: r = +0.7086 (R² = 50.2%)
CN-significant genes (n = 6,087): r = +0.8444 (R² = 71.3%)
Out-of-sample slope = 4.34 (predicted Δlog2[CN] ≈ 1.44)
Observed Differential Expression $\bar{Y}_{IBS} - \bar{Y}_{GBR}$ (Spanish vs British)
Within-Population EBV CN Slope $\hat{\beta}_{CN}$ Trained in 24 Non-IBS/GBR Populations
B
Targeted Out-of-Population Vector Alignment vs Global ANOVA
All 20,154 human genes (% variance R²)
CN-associated genes q < 0.05 (% variance R²)
Explained Between-Population Variance R² (%)
4.0%
8.3%
42.6%
61.9%
51.3%
72.1%
54.0%
73.8%
Global 25-df 26-Pop ANOVA
Same-Vial Spliced BZLF1
2019 WGS log2(EBV CN)
Multi-Parameter Viral Model